

Домашнее задание. RAG.

В этом домашнем задании вам предстоит создать туристический RAG, провести EDA и проявить креатив! :)

Вам дан набор данных, содержащий информацию о памятниках, архитектуре и других значимых местах. В данных есть:

- название достопримечательности;
- идентификатор на WikiData;
- город расположения;
- координаты долготы и широты;
- описание (извлеченное из WikiData);
- изображение в формате base64;
- сгенерированное описание изображения (генерация с помощью модели BLIP).

Некоторые из памятников могут встречаться несколько раз, но с различными изображениями, из которых не все могут быть качественными. Для компенсации этого даны текстовые описания.

Часть данных взята с Национальной технологической олимпиады студентов по CV. Базовое решение предполагает использование мультимодального ru-CLIP. Но в данном домашнем задании предлагается решить эту задачу творчески, используя NLP подход.

Установка и импорт библиотек

```
# !pip install -q gdown torch transformers transformers accelerate  
bitsandbytes langchain sentence-transformers faiss-gpu openpyxl  
datasets langchain-community ragatouille umap-learn
```

```
# !pip install --ignore-installed huggingface-hub==0.20.3  
# !pip uninstall huggingface-hub -y  
# !pip install huggingface-hub==0.20.3  
# !pip install --upgrade sentence-transformers  
# !pip install pacmap  
# !pip install -U bitsandbytes>=0.46.1  
# !pip uninstall langchain-community -y  
# !pip install langchain-community  
# !pip install --upgrade sentence-transformers  
# !pip install ragatouille
```

```
import warnings  
warnings.filterwarnings('ignore')
```

```
import os  
os.environ['LANGCHAIN_WARNINGS'] = 'false'
```

```
import gdown
```

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import torch
from tqdm.notebook import tqdm
from langchain.docstore.document import Document as LangchainDocument
from langchain.vectorstores import FAISS
from langchain_community.embeddings import HuggingFaceEmbeddings
from langchain_community.vectorstores.utils import DistanceStrategy

from transformers import AutoTokenizer, AutoModelForCausalLM,
BitsAndBytesConfig
import random
from langchain.text_splitter import SpacyTextSplitter
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.decomposition import PCA
import umap
from sklearn.metrics.pairwise import cosine_similarity
from FlagEmbedding import FlagReranker

device = 'cuda' if torch.cuda.is_available() else 'cpu'
print(f'Используется устройство: {device}')

Используется устройство: cuda

os.makedirs('content', exist_ok=True)

# !nvidia-smi

```

Загрузка данных (4 балла)

```

# url = 'https://drive.google.com/uc?id=1P1BsvI2jPN3fEqjc2YZxmQ-MTs22WVUk'
# output = 'file.csv'

# gdown.download(url, output, quiet=False)

# data = pd.read_csv('file.csv')

# Загрузка данных

file_id = '1P1BsvI2jPN3fEqjc2YZxmQ-MTs22WVUk'
output = 'content/file.csv'

if not os.path.exists(output):
    gdown.download(id=file_id, output=output, quiet=False)

data = pd.read_csv(output).drop('Unnamed: 0', axis=1, errors='ignore')

```

```
# data.sample(500)
```

```
data.sample(3)
```

	Name	WikiData	City
Lon \			
2691	Усадьба ротмистра Переяславцева	Q27890562	Екатеринбург
60.611946			
3981	Нижегородский Кремль	Q100270892	Нижний Новгород
44.005554			
5352	Локомотив	Q2503505	Нижний Новгород
43.929649			

	Lat	description \
2691	56.836205	Усадьба ротмистра Переяславцева
3981	56.327255	историческое здание в Нижнем Новгороде, Россия
5352	56.309647	многоцелевой стадион в Нижнем Новгороде, Россия

	image \
2691	/9j/4AAQSkZJRgABAQAAQABAAD/2wBDAAUEBAUEAwUFBA...
3981	/9j/4AAQSkZJRgABAQAAQABAAD/2wBDAAUEBAUEAwUFBA...
5352	/9j/4AAQSkZJRgABAQAAQABAAD/2wBDAAUEBAUEAwUFBA...

	en_txt
2691	two pictures of a building that has been destr...
3981	arafed view of a castle on a hill with a river...
5352	a close up of a group of different soccer team...

```
data['City'].unique()
```

```
<ArrowStringArray>
```

```
['Екатеринбург', 'Нижний Новгород', 'Владимир', 'Ярославль']
```

```
Length: 4, dtype: str
```

Проверим датасет на предмет пропусков

```
data.info()
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 14634 entries, 0 to 14633
```

```
Data columns (total 8 columns):
```

#	Column	Non-Null Count	Dtype
0	Name	14634 non-null	str
1	WikiData	12078 non-null	str
2	City	14634 non-null	str
3	Lon	14634 non-null	float64
4	Lat	14634 non-null	float64
5	description	12078 non-null	str
6	image	14634 non-null	str
7	en_txt	14634 non-null	str

```
dtypes: float64(2), str(6)
memory usage: 641.0 MB
```

```
data.isna().sum()
```

```
Name          0
WikiData      2556
City          0
Lon           0
Lat           0
description    2556
image         0
en_txt        0
dtype: int64
```

Удалим строки с пропусками в WikiData и проверим датасет еще раз на предмет пропусков

```
data = data.dropna(subset=['WikiData'])
```

```
data.isna().sum()
```

```
Name          0
WikiData       0
City          0
Lon           0
Lat           0
description     0
image         0
en_txt        0
dtype: int64
```

```
# sorted(data['WikiData'].unique(), reverse=True)
# sorted(data['description'].unique(), reverse=False)
sorted(data['en_txt'].unique(), reverse=True)
```

```
['прррмомонннннннннн',
'yellow house with a black roof and a white fence',
'yellow flowers in front of a brick building with a clock tower',
'yellow church with blue domes and a clock tower in a city',
'yellow church with a golden dome on a sunny day',
'yellow church with a golden dome on a cloudy day',
'yellow church with a golden dome and a steeple on a street corner',
'yellow church with a golden dome and a cross on top',
'yellow church with a blue dome and a cross on the front',
'yellow bus parked in front of a large yellow building with a green
roof',
'yellow building with white trim and windows on a sunny day',
'yellow building with white trim and windows on a street corner',
'yellow building with white trim and windows in winter time',
'yellow building with white trim and windows in snow covered area',
```

'yellow building with white trim and windows and a sign on the front',
'yellow building with white trim and windows',
'yellow building with white trim and columns on a snowy day',
'yellow building with white trim and columns',
'yellow building with white trim and arched windows on a sunny day',
'yellow building with white trim and arched windows in a city',
'yellow building with white trim and arched windows and a yellow car parked in front',
'yellow building with white pillars and a brick walkway in front of it',
'yellow building with white columns and pillars',
'yellow building with white columns and a green roof',
'yellow building with white columns and a balcony on a sunny day',
'yellow building with two towers and a yellow dome',
'yellow building with two towers and a golden dome',
'yellow building with two towers and a gold dome',
'yellow building with two towers and a clock on top of it',
'yellow building with green roof and white trim',
'yellow building with green roof and white columns',
'yellow building with green roof and green roofing',
'yellow building with green domes and a green roof',
'yellow building with columns and pillars on a street corner',
'yellow building with columns and a white door on a street',
'yellow building with columns and a green roof on a street corner',
'yellow building with columns and a green roof on a street',
'yellow building with columns and a brown door in front of it',
'yellow building with blue windows and a red door on a street',
'yellow building with arched windows and a green roof',
'yellow building with a white trim and a brown door',
'yellow building with a white roof and a green lawn',
'yellow building with a white and gold dome on top of it',
'yellow building with a steeple and a cross on top',
'yellow building with a sign that says metropolitan',
"yellow building with a sign that says bottroonirchich mark mark's method",
'yellow building with a sign on top of it that says hotel',
'yellow building with a sign on the front of it',
'yellow building with a sign and a sign post in front of it',
'yellow building with a red door and windows on a snowy day',
'yellow building with a red door and a clock on the front',
'yellow building with a lot of windows and a lot of cars',
'yellow building with a large flower garden in front of it',
'yellow building with a green roof and a white door',
'yellow building with a green roof and a white car parked in front of it',
'yellow building with a green roof and a clock tower',
'yellow building with a golden dome on top of it',
'yellow building with a golden dome and a horse drawn carriage',

'yellow building with a fountain in front of it and a tree in the middle',
'yellow building with a fountain in front of it and a fountain in the middle',
'yellow building with a fountain and statues in front of it',
'yellow building with a cross on the front of it',
'yellow building with a clock tower on top of it in a town',
'yellow building with a clock tower on top of it',
'yellow building with a clock tower and a walkway in front of it',
'yellow building with a clock tower and a flag on top',
'yellow building with a black roof and a fence in the snow',
'yellow building with a balcony and balconies on the top of it',
'yellow and white church with two golden domes and a cross on the top',
'yellow and white church with two gold domes and a green roof',
'yellow and white church with golden domes and a clock tower',
'yellow and white church with a green dome and a green roof',
'yellow and white church with a clock tower and a fence',
'yellow and white building with columns and windows on a street',
'yellow and white building with columns and a clock on the front',
'yellow and white building with a sign on the front of it',
'yellow and white building with a lot of windows and a fence',
'yellow and white building with a green roof and a green dome',
'yellow and white building with a green dome on top of it',
'yellow and white building with a green dome and a gold cross on top',
'yellow and white building with a clock tower and a woman walking by',
'yellow and white building with a clock on the front of it',
'yellow and red wooden house with a parking meter on the corner',
'yellow and green church with a steeple and a green roof',
'yellow and green building with two towers and a green roof',
'yellow and green building with a steeple and a clock tower',
'yellow and green building with a green trash can in front of it',
'yellow and green building with a green roof and a green roof',
'yellow and green building with a green dome and a blue roof',
'wrought iron fence with a green dome in the background',
'wrought iron fence surrounding a tree in front of a building',
'wrought iron fence surrounding a street with a building in the background',
'wrought iron fence surrounding a park with a building in the background',
'workers working on a large turbine in a factory',
'workers working on a large machine in a factory',
'woman taking a selfie in front of a mirror with her cell phone',
'woman standing on a stone wall overlooking a body of water',
'woman standing in front of a display case with a sculpture of a woman',
'woman sitting at a table with a laptop and a cup of coffee',

'woman sitting at a desk with a laptop and a book',
'woman in red jacket holding a large vase in front of a wall of musical instruments',
'woman in blue shirt using a machine to operate a machine',
'woman in black jacket listening to headphones in front of a black and white photograph',
'woman in a white top and floral skirt holding a box',
'woman holding a camera in front of a mirror with a blue light behind her',
'walkway in a park with trees and a statue in the distance',
'two women standing in a room with a table and pictures on the wall',
'two women looking at a model of the moon in a museum',
'two white lions laying on a roof in a zoo enclosure',
'two soccer players are playing soccer on a field with a crowd watching',
'two pictures of a tower with a clock on top of it',
'two pictures of a building that has been destroyed and a building that has been demolished',
'two pictures of a building and a car are hanging on a wall',
'two men sitting at a table with microphones in their hands',
'two men in uniform standing in front of a large sculpture',
'two men in suits sitting at a table signing papers',
'two men in hard hats looking at a turbine in a factory',
'two cheetah standing on rocks in a zoo enclosure',
'two boys in school uniforms are holding papers and looking at a tablet',
'trees with yellow leaves on them and a white church in the background',
'trees with yellow leaves in a park with a bench and a bench',
'trees with no leaves on them',
'trees lining the street in a park with a bench and a street light',
'trees lining the sidewalk in a park with people walking on the sidewalk',
'trees lining a path in a park with a lamp post',
'trees line the street in front of a white building with a clock tower',
'trees line the street in a park with a bench and a fire hydrant',
'trees line the edge of a grassy hill near a body of water',
'trees in the distance behind a grassy field with a dirt mound',
'trees in front of a building with cars parked in the parking lot',
'trees in front of a building with a clock tower in the middle',
'trees in front of a building with a clock tower',
'trees in front of a building with a clock on the top',
'trees in front of a building',
'trees in a park with a building in the background',
'trees in a park with a building and a clock tower',
'trees in a forest with a hill in the background',
'trees in a field with orange leaves and a cloudy sky',
'trees in a field with a bench and a bench in the middle',

'trees growing on the rocks in the woods near a blue sky',
'trees are turning colors in the fall and the water is still calm',
'trees are reflected in the water of a river in a city',
'trees are reflected in the water of a pond in front of a large house',
'trees are in the foreground of a large yellow building',
'trees are in the foreground of a city with a lake in the foreground',
'trees are in the foreground of a building with a street sign',
'trees are in the foreground of a building with a clock on a pedestal',
'trees are in the foreground and a monument in the background',
'trees are in the foreground and a clock tower is in the background',
'trees are in the foreground and a church in the background',
'trees are in the foreground and a building on the top of a hill',
'trees are in the foreground and a building is in the background',
'trees are in the foreground and a building in the background',
'trees are in the foreground',
'trees are in the background',
'trees are in front of the building',
'trees are in front of a white building with a clock tower',
'trees are in front of a red building with green domes',
'trees are in front of a large yellow building with a clock on the front',
'trees are in front of a church with a green roof',
'trees are in front of a church with a clock tower',
'trees are in front of a building with snow on the ground',
'trees are in front of a building with a snow covered sidewalk',
'trees are in front of a building with a sign on it',
'trees are in front of a building with a green roof',
'trees are in front of a building with a golden dome',
'trees are in front of a building with a clock tower',
'trees are in front of a building with a clock on the top',
'trees are growing on the side of the building',
'trees are growing on the side of a road in a park',
'trees are growing on the side of a hill near a castle',
'trees are growing on the hillside behind the castle',
'trees are growing on the hill behind the church',
'trees are growing in the park near a walkway and a building',
'trees and rocks in the woods with a bench and steps',
'trees and leaves are in the foreground of a pond and a bridge',
'trees and grass in a park with a clock tower in the background',
'trees and grass in a park with a brick path',
'trees and grass in a park with a bench and walkway',
'trees and grass in a field with a city in the background',
'trees and grass in a field with a blue sky in the background',
'trees and flowers in a park with benches and benches',
'trees and bushes surrounding a pond with a gazebo in the background',

'trees and bushes surrounding a pond and a church with a steeple',
'trees and bushes surrounding a lake with a building in the background',
'trees and bushes surround a walkway in a park with a statue',
'trees and bushes surround a park with a white building in the background',
'trees and bushes in front of a yellow building with a fountain',
'trees and bushes in front of a church with a steeple',
'trees and bushes in front of a building with a clothes line',
'trees and bushes in front of a brick wall with a clock tower',
'trees and bushes in a park with a bench and a building',
'trees and bushes in a field with a cloudy sky in the background',
'trees and a lake in the foreground with a church in the background',
'three priests standing in front of a building with a light pole',
'three men sitting at a table with microphones in front of them',
'three men in white lab coats are looking at a mirror',
'three men in military uniforms stand in front of a crowd',
'three men in military uniforms holding a certificate and a bouquet',
'three men in military uniforms are talking to each other',
'three men in military uniforms are looking at a cell phone',
'three girls sitting on a bed with a stuffed animal',
'three framed religious books on display in a glass case',
'three doctors standing in a hospital room with a surgical equipment',
'three doctors in white coats and masks stand in a room',
'they are two men standing in a room with a table full of clothes',
'they are all holding a child on their shoulders',
'there is a young girl standing on a bench in front of a building',
'there is a yellow taxi parked in front of a building',
'there is a yellow house with red trim and a red door',
'there is a yellow gate with a clock on top of it',
'there is a yellow gate with a clock on it in the middle of a street',
'there is a yellow fire hydrant sitting in front of a bar',
'there is a yellow church with two towers and a gate',
'there is a yellow car parked in front of a building',
'there is a yellow bus driving down the street in front of a building',
'there is a yellow building with two towers and a green roof',
'there is a yellow building with red trim and windows on the street',
'there is a yellow building with many windows and a tree',
'there is a yellow building with columns and a red stop light',
'there is a yellow building with a white window and a red stop sign',
'there is a yellow building with a white stripe on the side of it',
'there is a yellow building with a staircase going up to it',
'there is a yellow building with a sign that says ottohofen machachmeter',
'there is a yellow building with a sign on the top of it',
'there is a yellow building with a sign on the front of it',

'there is a yellow building with a red roof and a black fence',
'there is a yellow building with a red flag on the front',
'there is a yellow building with a green steeple and a clock tower',
'there is a yellow building with a green roof and a clock tower',
'there is a yellow building with a gate and a red fire hydrant',
'there is a yellow building with a clock tower in the middle of it',
'there is a yellow building with a clock on the top of it',
'there is a yellow building with a blue sign on the corner',
'there is a yellow building with a blue roof and a street sign',
'there is a yellow building with a blue dumpster in front of it',
'there is a yellow building with a black door and windows',
'there is a yellow building with a black door and a sign on the sidewalk',
'there is a yellow and white building with a sign on it',
'there is a yellow and white building with a green dome',
'there is a yellow and white building with a clock on top',
'there is a yellow and red building with a red roof',
'there is a yellow and green building with a green steeple',
'there is a yellow and green building with a green roof',
'there is a wooden wall with a shelf with spoons and spoons',
'there is a wooden structure with a steeple on top of it',
'there is a wooden structure with a cross on the top of it',
'there is a wooden path that is surrounded by trees',
'there is a wooden house with a snow covered roof and a window',
'there is a wooden house with a red roof and a tall building',
'there is a wooden house with a red roof and a red roof',
'there is a wooden house with a red roof and a large window',
'there is a wooden house with a red roof and a fence',
'there is a wooden house with a red roof and a brown roof',
'there is a wooden house with a fence and a gate',
'there is a wooden house with a door and a window',
'there is a wooden house with a bench in front of it',
'there is a wooden house with a balcony and stairs in the yard',
'there is a wooden fence and a wooden house in the background',
'there is a wooden deck with stairs leading to a house',
'there is a wooden deck with benches and a water slide',
'there is a wooden deck with benches and a fire pit',
'there is a wooden cross in front of a wooden building',
'there is a wooden church with two steeples on top of it',
'there is a wooden cart with a wooden seat in a museum',
'there is a wooden building with graffiti on it on the side of the street',
'there is a wooden building with a wooden fence and a wooden fence',
'there is a wooden building with a windmill on the roof',
'there is a wooden building with a steeple on the top of it',
'there is a wooden building with a steeple on the top',
'there is a wooden building with a sign on the front of it',
'there is a wooden building with a sign on it',
'there is a wooden building with a red roof and a red roof',

'there is a wooden building with a metal roof and a street sign',
'there is a wooden building with a lot of windows on the side of it',
'there is a wooden building with a large window next to a sidewalk',
'there is a wooden building with a green roof and a wooden fence',
'there is a wooden building with a brown roof and white trim',
'there is a wooden bed in a room with a canopy',
'there is a woman walking down the street with a stroller',
'there is a woman walking down the street in the snow',
'there is a woman walking down the street in front of a building',
'there is a woman walking down a path in a park',
'there is a woman walking down a path in a garden',
'there is a woman that is working on a piece of wood',
'there is a woman that is standing in a cemetery with a bat',
'there is a woman that is sitting on a rock by the water',
'there is a woman that is sitting down with a cell phone',
'there is a woman that is holding a pillow in her hand',
'there is a woman standing on a path in the woods',
'there is a woman standing next to a horse in a barn',
'there is a woman standing in front of a yellow building',
'there is a woman standing in front of a train with a wagon',
'there is a woman standing in front of a projector screen',
'there is a woman standing in front of a painting on a easel',
'there is a woman standing in front of a monument in a city',
'there is a woman standing in front of a building with a fountain',
'there is a woman standing in front of a building with a clock on
it',
'there is a woman standing in a theatre with a microphone',
'there is a woman standing in a room with a lot of art',
'there is a woman standing in a room with a dog',
'there is a woman standing in a room with a chandelier',
'there is a woman standing in a park with a cell phone',
'there is a woman sitting on a couch with a laptop',
'there is a woman sitting on a bench in front of a television',
'there is a woman sitting on a bench in a park',
'there is a woman sitting at a table with a painting on it',
'there is a woman sitting at a table with a laptop',
'there is a woman crossing the street in front of a building',
'there is a woman and a child walking down a path',
'there is a woman and a child standing in the snow',
'there is a window on the side of a building with a plaque on it',
'there is a white truck parked in front of a building',
'there is a white tower with a cross on top in a park',
'there is a white tower with a clock on it in a park',
'there is a white structure in the middle of a lake',
'there is a white monument in the middle of a snowy park',
'there is a white lion standing in a zoo enclosure',
'there is a white house with a wooden door and a fence',
'there is a white house with a clock tower in the background',
'there is a white house with a black roof and a white fence',

'there is a white gazebo with a green roof on a brick walkway',
'there is a white gazebo with a green roof and a man standing on the side',
'there is a white gazebo with a green roof and a black fence',
'there is a white gazebo with a black roof on a hill',
'there is a white gazebo sitting on the side of a road',
'there is a white gazebo sitting on the shore of a lake',
'there is a white gazebo on the side of a road',
'there is a white gazebo in the middle of a grassy area',
'there is a white gate with a sign that says fort town gardens',
'there is a white fence with a gate and a tree',
'there is a white fence and a white house on a hill',
'there is a white clock tower with a bell on top of it',
'there is a white clock tower in the middle of a snowy park',
'there is a white clock tower in the middle of a park',
'there is a white church with two towers and a steeple',
'there is a white church with green domes on the side of the road',
'there is a white church with green domes on a grassy hill',
'there is a white church with green domes and a green roof',
'there is a white church with green domes and a fence',
'there is a white church with blue domes on the side of the road',
'there is a white church with blue domes on a cloudy day',
'there is a white church with blue domes and a blue roof',
'there is a white church with a steeple and a tree in front',
'there is a white church with a steeple and a green roof',
'there is a white church with a steeple and a clock tower',
'there is a white church with a steeple and a blue roof',
'there is a white church with a heart shaped statue in the middle',
'there is a white church with a green steeple on the side of it',
'there is a white church with a green steeple in the middle of a snowy field',
'there is a white church with a green steeple and a green dome',
'there is a white church with a green roof on a beach',
'there is a white church with a green roof and a steeple',
'there is a white church with a green roof and a blue dome',
'there is a white church with a golden dome on top',
'there is a white church with a gold steeple and a clock tower',
'there is a white church with a gold dome and a clock tower',
'there is a white church with a cross on the top of it',
'there is a white church with a clock tower in the middle of the park',
'there is a white church with a blue roof and a bell tower',
'there is a white church tower with a clock on it',
'there is a white church tower on the shore of a lake',
'there is a white church on the shore of a lake',
'there is a white building with two towers and a clock on the top',
'there is a white building with two towers and a blue roof',
'there is a white building with three towers and a blue roof',
'there is a white building with green roof and a green roof',

'there is a white building with green domes on the top of it',
'there is a white building with green domes on it',
'there is a white building with green domes and a clock tower',
'there is a white building with columns and a car parked on the street',
'there is a white building with blue domes on top of it',
'there is a white building with blue domes on it',
'there is a white building with blue domes in the snow',
'there is a white building with blue domes in the middle of a forest',
'there is a white building with blue domes and a yellow roof',
'there is a white building with blue domes and a fence',
'there is a white building with blue domes and a clock tower',
'there is a white building with a yellow awning on the front',
'there is a white building with a wooden door and a window',
'there is a white building with a tower on top of it',
'there is a white building with a tower and a clock tower',
'there is a white building with a steeple on top of it',
'there is a white building with a steeple on the top of it',
'there is a white building with a steeple and a steeple on the side',
'there is a white building with a steeple and a steeple on the front',
'there is a white building with a steeple and a clock tower',
'there is a white building with a staircase going up to it',
'there is a white building with a sign on the front of it',
'there is a white building with a green roof on a hill',
'there is a white building with a green roof next to a body of water',
'there is a white building with a green roof and a steeple',
'there is a white building with a green roof and a staircase',
'there is a white building with a green roof and a green roof',
'there is a white building with a green roof and a clock tower',
'there is a white building with a green roof and a blue roof',
'there is a white building with a green dome on top',
'there is a white building with a green dome and a green roof',
'there is a white building with a green dome and a gold cross on top',
'there is a white building with a golden dome on top',
'there is a white building with a gold dome on top',
'there is a white building with a gold dome in the middle of a garden',
'there is a white building with a gold dome and a bicycle parked in front of it',
'there is a white building with a fence and a building with a blue roof',
'there is a white building with a dome on top of it',
'there is a white building with a cross on top of it',
'there is a white building with a clock tower on the corner',
'there is a white building with a clock tower in the background',

'there is a white building with a clock on the top of it',
'there is a white building with a clock on the front of it',
'there is a white building with a brown roof on a street',
'there is a white building with a brown roof and a white tower',
'there is a white building with a brown roof and a red roof',
'there is a white building with a brown roof and a clock tower',
'there is a white building with a brown roof and a car parked in front of it',
'there is a white building with a brown roof and a brown door',
'there is a white building with a brown door and steps',
'there is a white building with a brown door and a brown fence',
'there is a white building with a blue steeple and a clock tower',
'there is a white building with a blue sign on it',
'there is a white building with a blue roof on a hill',
'there is a white building with a blue roof and a steeple',
'there is a white building with a blue roof and a green hedge',
'there is a white building with a blue roof and a blue dome',
'there is a white building with a blue dome on top',
'there is a white building with a black roof and a yellow top',
'there is a white building with a black roof and a white dome',
'there is a white building with a black roof and a street sign',
'there is a white building with a black roof and a staircase',
'there is a white building with a black roof and a clock tower',
'there is a white building with a black fence and a street sign',
'there is a white building with a black fence and a street light',
'there is a white building with a black fence and a body of water',
'there is a white building with a black and white striped pole',
'there is a white building with a black and white pole in front of it',
'there is a white building with a balcony on the top of it',
'there is a white building with a balcony and a balcony on the second floor',
'there is a white building on the side of the water',
'there is a white building on the hill with a car parked in front',
'there is a white bridge that is over a road',
'there is a white bench sitting in front of a yellow building',
'there is a white archway with a sign that says thomas e thomas memorial',
'there is a wall with a lot of pictures on it',
'there is a wall with a clock and a bunch of bells',
'there is a walkway with a bench and a bike on it',
'there is a walkway that leads to a playground with a slide',
'there is a walkway that leads to a park with a playground',
'there is a walkway that leads to a monument in a park',
'there is a walkway that leads to a monument in a cemetery',
'there is a walkway that leads to a gazebo in the park',
'there is a walkway that has a bench on it in the middle of a park',
'there is a walkway in the middle of a park with trees',
'there is a vintage photo of a woman in a park',
'there is a view of a town from a hill with a few people',

'there is a view of a park with a skateboard ramp and a lake',
'there is a view of a garden from a balcony of a house',
'there is a view of a city from a high up',
'there is a view of a church through a doorway of a building',
'there is a very tall building with a clock tower on top',
'there is a very ornate altar with flowers and a painting on the wall',
'there is a very old wooden house with a lot of windows',
'there is a very old house with a red trim on it',
'there is a very old building with a lot of windows',
'there is a vase with a floral design on it in a room',
'there is a typewriter and a book on a desk',
'there is a truck that is driving down the street',
'there is a truck parked on the side of the road in front of a building',
'there is a truck parked in front of a building with a truck in front of it',
'there is a truck parked in front of a building on the street',
'there is a tree that is in front of a building',
'there is a tree that is growing out of a rock in the woods',
'there is a tree that has fallen down in the park',
'there is a tree that has fallen down in front of a building',
'there is a tree that has been cut down in the woods',
'there is a tree that has been cut down and is in the yard',
'there is a tree in front of a building with snow on the ground',
'there is a train track going through a small town with a building in the background',
'there is a train that is sitting on the tracks in the room',
'there is a train that is parked in front of a building',
'there is a train that is going over a bridge over water',
'there is a train that is going over a bridge over a river',
'there is a train that is going down the tracks in the city',
'there is a train that is going down the tracks in front of a building',
'there is a train that is going down the tracks by the water',
'there is a train that is going by a building that is being demolished',
'there is a train that is going by a building on the tracks',
'there is a train that is going around a circle of tracks',
'there is a traffic light on the corner of a street',
'there is a tower with a spiral staircase in front of a building',
'there is a tower with a spiral staircase and a staircase leading to it',
'there is a tower with a clock on top of it in a city',
'there is a tower with a clock on top of it',
'there is a tiger walking in the snow near a building',
'there is a tiger that is sitting on a wooden platform',
'there is a tiger that is laying on a table in the grass',
'there is a tiger laying on a wooden platform in a zoo',

'there is a tiger laying on a platform in a zoo',
'there is a television set with a picture of a man on it',
'there is a teddy bear standing in the middle of a flower garden',
'there is a teddy bear sitting in a corner of a room',
'there is a teddy bear and a bear doll in a toy house',
'there is a tank that is sitting on the ground in front of a building',
'there is a tank that is sitting on a pedestal in the middle of a street',
'there is a tall white tower with a face on it',
'there is a tall white clock tower in front of a building',
'there is a tall tower with a steeple on top of it',
'there is a tall tower with a staircase going up to it',
'there is a tall tower with a gold top on a building',
'there is a tall tower with a flag on top of it',
'there is a tall tower with a face on it in the middle of a field',
'there is a tall tower with a clock on top of it',
'there is a tall tower with a clock on it on a hill',
'there is a tall tower with a clock on it next to a fence',
'there is a tall tower with a clock on it in the middle of a town',
'there is a tall tower with a clock on it in the middle of a park',
'there is a tall tower with a clock on it in the middle of a city',
'there is a tall tower with a clock on it in a town',
'there is a tall tower with a clock on it in a park',
'there is a tall statue with a clock on top of it',
'there is a tall red brick building with a clock on top',
'there is a tall monument with a gold angel on top',
'there is a tall monument with a clock on top of it',
'there is a tall monument in the middle of a park',
'there is a tall clock tower in the middle of a park',
'there is a tall building with two windows and a clock on it',
'there is a tall building with many windows on it',
'there is a tall building with a tower on the corner',
'there is a tall building with a steeple on the top',
'there is a tall building with a steeple and a bell tower',
'there is a tall building with a sign on the side of it',
'there is a tall building with a sign on the front of it',
'there is a tall building with a red roof and a tall building with a red roof',
'there is a tall building with a lot of windows on it',
'there is a tall building with a golden dome on top of it',
'there is a tall building with a clock tower on top of it',
'there is a tall building with a clock tower on the top',
'there is a tall building with a clock tower in the middle of it',
'there is a tall building with a clock tower and a cross on it',
'there is a tall building with a clock on the top of it',
'there is a tall building with a clock on the front of it',
'there is a tall building with a balcony on the corner of the street',

'there is a tall building with a balcony and a balcony on the second floor',
'there is a tall brick wall with a clock tower on it',
'there is a tall brick tower with a steeple on top of it',
'there is a tall brick tower with a green roof on a street corner',
'there is a tall brick tower with a green roof on a street',
'there is a tall brick tower with a green roof on a hill',
'there is a tall brick tower with a green roof and a staircase',
'there is a tall brick tower with a green roof and a clock on top',
'there is a tall brick tower with a green roof and a blue toilet',
'there is a tall brick tower with a green roof',
'there is a tall brick tower with a cross on top',
'there is a tall brick tower with a clock on top of it',
'there is a tall brick tower with a clock on top',
'there is a tall brick tower with a clock on it',
'there is a tall brick building with a steeple on top',
'there is a tall brick building with a steeple on the top',
'there is a tall brick building with a green roof on a street corner',
'there is a tall brick building with a green roof on a street',
'there is a tall brick building with a green roof and a clock on top',
'there is a tall brick building with a green roof',
'there is a tall brick building with a clock tower on top',
'there is a tall brick building with a clock on the top',
'there is a tall brick building with a clock on the front',
'there is a table with red chairs and a large painting on the wall',
'there is a table with chairs and a couch in a room',
'there is a table with a white table cloth on it',
'there is a table with a white table cloth and chairs',
'there is a table with a white table cloth and a white table cloth',
'there is a table with a white table cloth and a chair',
'there is a table with a white table cloth and a black piano',
'there is a table with a vase on it in a room',
'there is a table with a vase of flowers on it',
'there is a table with a vase and a vase on it',
'there is a table with a variety of rocks and a lizard',
'there is a table with a table cloth and a basket on it',
'there is a table with a red table cloth and chairs in a room',
'there is a table with a mirror and a chair in a room',
'there is a table with a glass top in a room',
'there is a table with a computer and a book on it',
'there is a table with a book on it in a room',
'there is a table with a book and a vase on it',
'there is a table with a book and a lamp in a room',
'there is a table with a book and a candle on it',
'there is a table with a blue table cloth and a chair',
'there is a street with a white building and a white truck',
'there is a street with a red brick building and a white line',

'there is a street with a lot of buildings on it',
'there is a street with a fence and a building with a clock on it',
'there is a street with a car driving down it and a building in the background',
'there is a street with a building and a tree in the middle',
'there is a street with a building and a street light',
'there is a street with a building and a clock tower in the background',
'there is a street with a building and a clock on it',
'there is a street with a building and a car parked on the side',
'there is a street view of a building with a clock tower',
'there is a street sign that is on the corner of a street',
'there is a street sign that is on the corner of a building',
'there is a street sign on the sidewalk next to a building',
'there is a street sign on the corner of a street',
'there is a street scene with a car and a building',
'there is a street light that is on the side of a building',
'there is a street light on the side of the road',
'there is a street light on the corner of a street in front of a building',
'there is a street light in the middle of a park',
'there is a stone with a plaque on it in the middle of a garden',
'there is a stone with a hole in it in the snow',
'there is a stone with a cross on it and a poem on it',
'there is a stone statue on the ground next to a trash can',
'there is a stone staircase going up to a rock formation',
'there is a stone staircase going up a hill in the woods',
'there is a stone seating area in the middle of a forest',
'there is a stone sculpture of a man on a brick path',
'there is a stone path leading to a memorial in a park',
'there is a stone monument with a clock on it in a park',
'there is a stone monument in the middle of a park',
'there is a stone grave with a cross on it in a cemetery',
'there is a stone bridge with graffiti on it over a frozen river',
'there is a stone bridge that is going over a small bridge',
'there is a stone bridge over a river with a stone arch',
'there is a statue of two people holding flags in front of a building',
'there is a statue of two men holding flags in a park',
'there is a statue of three men fighting each other',
'there is a statue of an eagle on top of a pole',
'there is a statue of a woman with a wreath on it',
'there is a statue of a woman sitting on a pile of luggage',
'there is a statue of a woman sitting on a bench in a garden',
'there is a statue of a woman in a park with a lamp post',
'there is a statue of a woman holding a flag in the snow',
'there is a statue of a woman holding a flag in a park',
'there is a statue of a woman holding a cross in a cemetery',
'there is a statue of a woman holding a child near a church',

'there is a statue of a woman holding a book in a snowy park',
'there is a statue of a woman holding a book in a park',
'there is a statue of a woman holding a bird in the snow',
'there is a statue of a woman holding a baby in a cemetery',
'there is a statue of a soldier standing next to a table',
'there is a statue of a man with a wreath on it',
'there is a statue of a man with a red hat and a red flower',
'there is a statue of a man with a horse head on it',
'there is a statue of a man with a hat and a jacket',
'there is a statue of a man with a hat and a coat',
'there is a statue of a man with a crown on it',
'there is a statue of a man with a cross and a cross on it',
'there is a statue of a man with a child and a man with a dog',
'there is a statue of a man with a bust on a pedestal',
'there is a statue of a man with a book on a street',
'there is a statue of a man with a beard on a pedestal',
'there is a statue of a man with a beard in the snow',
'there is a statue of a man with a beard in front of a building',
'there is a statue of a man with a beard in a park',
'there is a statue of a man with a beard in a cemetery',
'there is a statue of a man standing on top of a box',
'there is a statue of a man standing on a rock in the snow',
'there is a statue of a man standing on a rock in the middle of a park',
'there is a statue of a man standing on a rock in front of a building',
'there is a statue of a man standing on a rock in a park',
'there is a statue of a man standing on a rock',
'there is a statue of a man standing on a pedestal in the middle of a street',
'there is a statue of a man standing on a pedestal in front of a building',
'there is a statue of a man standing on a pedestal in a park',
'there is a statue of a man standing on a pedestal in a city',
'there is a statue of a man standing on a pedestal',
'there is a statue of a man standing on a bench',
'there is a statue of a man standing next to a building',
'there is a statue of a man standing in the snow',
'there is a statue of a man standing in the middle of a street',
'there is a statue of a man standing in the middle of a snowy park',
'there is a statue of a man standing in the middle of a park',
'there is a statue of a man standing in front of a wall',
'there is a statue of a man standing in front of a tree',
'there is a statue of a man standing in front of a sign',
'there is a statue of a man standing in front of a monument',
'there is a statue of a man standing in front of a forest',
'there is a statue of a man standing in front of a building',
'there is a statue of a man standing in front of a bench',
'there is a statue of a man standing in a square',

'there is a statue of a man standing in a park',
'there is a statue of a man standing in a fountain',
'there is a statue of a man sitting on a pedestal in the park',
'there is a statue of a man sitting on a pedestal in the middle of a street',
'there is a statue of a man sitting on a pedestal in the middle of a park',
'there is a statue of a man sitting on a pedestal in front of a building',
'there is a statue of a man sitting on a pedestal in a park',
'there is a statue of a man sitting on a pedestal',
'there is a statue of a man sitting on a grave',
'there is a statue of a man sitting on a chair in the woods',
'there is a statue of a man sitting on a bench next to a wall',
'there is a statue of a man sitting on a bench in the park',
'there is a statue of a man sitting on a bench in the grass',
'there is a statue of a man sitting on a bench in front of a monument',
'there is a statue of a man sitting on a bench in front of a building',
'there is a statue of a man sitting on a bench in a park',
'there is a statue of a man sitting on a bench',
'there is a statue of a man riding a horse with a man on it',
'there is a statue of a man riding a horse with a helmet on',
'there is a statue of a man riding a horse on a pedestal',
'there is a statue of a man riding a horse in the park',
'there is a statue of a man riding a horse in front of a building',
'there is a statue of a man on the side of a building',
'there is a statue of a man on a staircase in a building',
'there is a statue of a man on a skateboard in the middle of a street',
'there is a statue of a man on a rock in front of a building',
'there is a statue of a man on a pole in front of a building',
'there is a statue of a man on a pedestal with a smoke stack in the background',
'there is a statue of a man on a pedestal in the middle of a park',
'there is a statue of a man on a pedestal in front of a stop sign',
'there is a statue of a man on a pedestal in front of a building',
'there is a statue of a man on a pedestal in a park',
'there is a statue of a man on a horse in front of a building',
'there is a statue of a man on a bench in front of a building',
'there is a statue of a man in front of a building',
'there is a statue of a man in a suit on a pedestal',
'there is a statue of a man in a suit in front of a building',
'there is a statue of a man in a suit in a park',
'there is a statue of a man in a suit in a flower bed',
'there is a statue of a man in a suit and tie',
'there is a statue of a man in a park with trees',
'there is a statue of a man in a park with flowers',

'there is a statue of a man in a medieval armor room',
'there is a statue of a man in a coat on a pedestal',
'there is a statue of a man in a coat and a flag',
'there is a statue of a man in a coat and a building in the background',
'there is a statue of a man in a cage in a room',
'there is a statue of a man in a bust on a square',
'there is a statue of a man holding a woman on his back',
'there is a statue of a man holding a violin in a park',
'there is a statue of a man holding a sword in front of a building',
'there is a statue of a man holding a sword in a square',
'there is a statue of a man holding a sword in a plaza',
'there is a statue of a man holding a sword in a park',
'there is a statue of a man holding a knife in front of a building',
'there is a statue of a man holding a hat in front of a tree',
'there is a statue of a man holding a hat in a park',
'there is a statue of a man holding a gun in front of a building',
'there is a statue of a man holding a gun in a plaza',
'there is a statue of a man holding a gun in a park',
'there is a statue of a man holding a gun in a city square',
'there is a statue of a man holding a guitar in front of a building',
'there is a statue of a man holding a globe in the snow',
'there is a statue of a man holding a flag on a pedestal',
'there is a statue of a man holding a flag next to a clock tower',
'there is a statue of a man holding a flag in the snow',
'there is a statue of a man holding a flag in front of a building',
'there is a statue of a man holding a flag in a park',
'there is a statue of a man holding a flag and a woman holding a flag',
'there is a statue of a man holding a flag and a clock tower',
'there is a statue of a man holding a cigarette in front of a building',
'there is a statue of a man holding a child in front of a church',
'there is a statue of a man holding a child in a park',
'there is a statue of a man holding a briefcase in front of a building',
'there is a statue of a man holding a book in a plaza',
'there is a statue of a man holding a book in a park',
'there is a statue of a man holding a bird in the snow',
'there is a statue of a man holding a baseball bat in front of a building',
'there is a statue of a man holding a baseball bat in a room',
'there is a statue of a man holding a baseball bat in a park',
'there is a statue of a man holding a baseball bat',
'there is a statue of a man holding a baby in a park',
'there is a statue of a man holding a baby in a city square',
'there is a statue of a man and a woman on a train track',
'there is a statue of a man and a woman on a pedestal',
'there is a statue of a man and a woman on a horse',

'there is a statue of a man and a woman on a bridge',
'there is a statue of a man and a woman on a bench',
'there is a statue of a man and a woman in front of a building',
'there is a statue of a man and a woman in a park',
'there is a statue of a man and a woman holding flags',
'there is a statue of a man and a woman holding a flag',
'there is a statue of a man and a horse on a trailer',
'there is a statue of a man and a horse in front of a church',
'there is a statue of a lion with a shield on it',
'there is a statue of a horse on a pedestal in a park',
'there is a statue of a gun on display in front of a brick building',
'there is a statue of a dog on a pedestal in the snow',
'there is a statue of a cat in front of a building',
'there is a statue of a bird on top of a column',
'there is a statue of a bird in the middle of a park',
'there is a statue of a bear that is on a pedestal',
'there is a statue of a bear that is on a brick walkway',
'there is a statue of a bear in front of a building',
'there is a statue of a bear and a dog in a park',
'there is a statue in the middle of a walkway in front of a building',
'there is a statue in the middle of a street with a red flower bed',
'there is a statue in the middle of a square with a clock tower in the background',
'there is a statue in the middle of a park with trees',
'there is a statue in the middle of a park with people walking by',
'there is a statue in the middle of a park with benches',
'there is a statue in the middle of a park with a tree',
'there is a statue in the middle of a park with a fountain',
'there is a statue in the middle of a park with a clock on it',
'there is a statue in the middle of a park with a bench',
'there is a statue in the middle of a park surrounded by trees',
'there is a statue in the middle of a flower bed in front of a building',
'there is a statue in front of a building with a clock on it',
'there is a staircase with a railing and a vase on the table',
'there is a staircase leading to a second floor with pictures on the wall',
'there is a staircase leading to a large room with a red wall',
'there is a staircase in a house with a chandelier and pictures on the wall',
'there is a staircase in a building with a red fire hydrant',
'there is a staircase in a building with a painting on the wall',
'there is a staircase in a building with a chandelier and a chandelier',
'there is a spoon and spoon rest in a box on a table',
'there is a spiral staircase with a glass railing and a clock on the wall',
'there is a spiral staircase with a chandelier and a table',

'there is a snow covered wall with a tower in the background',
'there is a snow covered street with a building and a snowboard',
'there is a snow covered house with a tree in front of it',
'there is a snow covered field with a clock tower in the background',
'there is a snow covered building with a sign on it',
'there is a small wooden house with a wooden fence and a wooden gate',
'there is a small wooden house with a tree in the yard',
'there is a small wooden house with a red roof and a green fence',
'there is a small wooden house with a porch and a fence',
'there is a small wooden house with a porch and a bench',
'there is a small wooden house with a metal roof',
'there is a small wooden house with a green roof',
'there is a small wooden house with a gate and a fence',
'there is a small wooden house with a brown roof',
'there is a small wooden house with a balcony on the side of the street',
'there is a small wooden house on the corner of a street',
'there is a small wooden house in the snow with a tree',
'there is a small wooden house in the middle of a field',
'there is a small wooden church with two steeples on top of it',
'there is a small wooden church in the middle of a grassy field',
'there is a small wooden church in the middle of a field',
'there is a small wooden building with two towers in the middle of the woods',
'there is a small wooden building with two domes on top of it',
'there is a small wooden building with three spires on top of it',
'there is a small wooden building with a white roof',
'there is a small wooden building with a white fence and a clock tower',
'there is a small wooden building with a steeple on top',
'there is a small wooden building with a steeple on the top',
'there is a small wooden building with a sign on the front',
'there is a small wooden building with a sign on it',
'there is a small wooden building with a red roof and a red tin roof',
'there is a small wooden building with a red light on the corner',
'there is a small wooden building with a metal roof',
'there is a small wooden building with a dome on top',
'there is a small wooden building with a cross on the front',
'there is a small wooden building with a clock tower on top',
'there is a small wooden building with a clock on top',
'there is a small wooden building with a clock on the top of it',
'there is a small wooden building in the woods with a snow covered ground',
'there is a small white gazebo on the side of a lake',
'there is a small white church with blue domes on it',
'there is a small white church with a green roof',
'there is a small white building with blue domes on it',

'there is a small white building with a green roof on a hill',
'there is a small white building with a brown roof on a hill',
'there is a small white building with a bell tower in the middle of a field',
'there is a small white building in the middle of a park',
'there is a small white and orange building with two towers',
'there is a small tree that is growing out of a cement step',
'there is a small table with a cross on it in a church',
'there is a small river running through a city with a building on the side',
'there is a small red brick building with a clock on the top',
'there is a small red and white house with a green door',
'there is a small pond in front of a building with a bridge',
'there is a small playground in front of a large white building',
'there is a small plane that is parked in the grass',
'there is a small plane that is on top of a building',
'there is a small plane on top of a building in a city',
'there is a small pink building with a steeple on the top',
'there is a small pink building with a green roof and a steeple',
'there is a small monument with a star on it in a cemetery',
'there is a small monument in the middle of a street',
'there is a small log cabin with a sign on the front',
'there is a small log cabin with a red roof and a red door',
'there is a small house with a wooden fence in front of it',
'there is a small house with a tree in front of it',
'there is a small house with a red roof and a white fence',
'there is a small house with a red roof and a white door',
'there is a small house with a red roof and a fence',
'there is a small house with a porch and a covered patio',
'there is a small house with a porch and a bench in the yard',
'there is a small house with a ladder in the front yard',
'there is a small house with a hammock on the porch',
'there is a small house with a green roof and a red brick building',
'there is a small house with a green roof and a black gate',
'there is a small house with a blue roof and a black roof',
'there is a small house on the corner of a street',
'there is a small grave with a large head and a small grave with a large head',
'there is a small garden with a lot of flowers and trees',
'there is a small dog walking in front of a house',
'there is a small dining room with a couch and a table',
'there is a small church with two steeples on the top of it',
'there is a small church with two steeples on the roof',
'there is a small church with a steeple on the top of it',
'there is a small church with a steeple on the top',
'there is a small church with a steeple on the roof',
'there is a small church with a steeple and a steeple on the side',
'there is a small church with a steeple and a steeple on the roof',
'there is a small church with a steeple and a green roof',

'there is a small church with a steeple and a cross on top',
'there is a small church with a steeple and a cross on the top',
'there is a small church with a rainbow hose in front of it',
'there is a small church with a green steeple on the top',
'there is a small church with a green roof and a steeple',
'there is a small church with a cross on top of it',
'there is a small church with a cross on the top of it',
'there is a small church with a bell tower in the middle of a street',
'there is a small church on the corner of a street',
'there is a small church in the middle of a small town',
'there is a small cabin in the woods with a porch',
'there is a small building with a yellow dome on top',
'there is a small building with a tower on the top of it',
'there is a small building with a snow covered roof on the street',
'there is a small building with a small entrance in front of it',
'there is a small building with a sign on the front of it',
'there is a small building with a sign on it that says the museum',
'there is a small building with a red door and a window',
'there is a small building with a metal roof and a wooden fence',
'there is a small building with a metal roof and a sign on the front',
'there is a small building with a green roof and a tower',
'there is a small building with a golden dome on top',
'there is a small building with a flower pot on the side of it',
'there is a small building with a dome on the top of it',
'there is a small building with a cross on top of it',
'there is a small building with a clock tower on the side of it',
'there is a small building with a clock on the top of it',
'there is a small building with a clock on the side of it',
'there is a small brick structure with a dome on top',
'there is a small brick building with a steeple on the top',
'there is a small brick building with a green roof and a steeple',
'there is a small brick building with a green roof and a green door',
'there is a small brick building with a green roof',
'there is a small brick building with a clock tower on top',
'there is a small boy riding a toy car in front of a church',
'there is a small blue gazebo in the middle of a garden',
'there is a small black building with two steeples on top',
'there is a small bench in the grass next to a tree',
'there is a small bedroom with a bed and desk in it',
'there is a small bed in a small room with a window',
'there is a small bathroom with a sink and a window',
'there is a skateboarder riding on a ramp in front of a building',
'there is a skateboarder riding down a ramp outside of a building',
'there is a sign that says the hotel is hanging from the side of a building',
'there is a sign that says the city of st petersburg',
'there is a sign that says russian language on it',
'there is a sign that says restaurant in russian on the side of a

building',
'there is a sign that says movini in russian on a building',
'there is a sign that says hopewood in front of a path',
'there is a sign that says georgetown street wrong 3',
'there is a sign that is on the side of the street',
'there is a sign that is on the side of a tree',
'there is a sign that is on the side of a dirt road',
'there is a sign that is in the snow by the trees',
'there is a sign that is in the middle of the forest',
'there is a sign that is in the grass near a tree',
'there is a sign that is in the grass near a path',
'there is a sign that is in the grass by a tree',
'there is a sign that is in front of a building',
'there is a sign on the wall that says befuha ciara teporm',
'there is a sign on the wall of a building that says occibik kyhila',
'there is a sign on the wall of a building that says it is a time',
'there is a sign on the wall of a building that says house of
justice',
'there is a sign on the wall of a building that says',
'there is a sign on the sidewalk that says that there is a building',
'there is a sign on the side of a building that says, hotel
maritana',
'there is a sign on the side of a building that says,',
'there is a sign on the side of a building that says old city',
'there is a sign on the side of a building that says miksish
mikkish',
'there is a sign on the side of a building that says korphyc',
'there is a sign on the side of a building that says',
'there is a sign on the side of a bridge that says montaupoy hotel',
'there is a sign on the path that says the trail is not open',
'there is a sign on the fence that says, 'the cemetery is closed ''',
'there is a sign on the door of a building that says occinik',
'there is a sign on the building that says russian language',
'there is a sign on the building that says red lion',
'there is a sidewalk that has a sign on it that says the street is
narrow',
'there is a sheep that is walking in the snow',
'there is a sculpture of a man sitting on a chair in a building',
'there is a sculpture of a man on a horse in a museum',
'there is a sculpture in the middle of a room with paintings on the
wall',
'there is a sculpture in the middle of a brick walkway',
'there is a room with tables and chairs and a clock on the wall',
'there is a room with a window and a chair in it',
'there is a room with a wall with pictures and a coat rack',
'there is a room with a table, chairs, couches and paintings',
'there is a room with a table, chairs, and paintings on the walls',
'there is a room with a table, chairs, and a plant in it',
'there is a room with a table, chairs, and a mirror',

```
'there is a room with a table, chairs, and a dresser',
'there is a room with a table, chairs, and a couch',
'there is a room with a table, chairs, and a clock',
'there is a room with a table, chairs, and a cabinet',
'there is a room with a table, chair, and a clock',
'there is a room with a table, a clock, and a vase',
'there is a room with a table and pictures on the wall',
'there is a room with a table and chairs in it',
'there is a room with a table and chairs and a painting on the wall',
'there is a room with a table and chairs and a chandelier',
'there is a room with a table and a window with curtains',
'there is a room with a table and a wall with pictures',
'there is a room with a table and a lot of food on it',
'there is a room with a table and a large display case',
'there is a room with a table and a chair in it',
'there is a room with a stove, a table, and chairs',
...]
```

В колонке 'en_txt' замечено значение 'прррмомонннннннннн', заменим его на моду

```
mask = data['en_txt'] == 'прррмомонннннннннн'
name_field = data[mask]['Name'].iloc[0]
print(name_field)

mode_value = data[data['Name'] == name_field]['en_txt'].mode()[0]
print(mode_value)

data.loc[mask, 'en_txt'] = mode_value
```

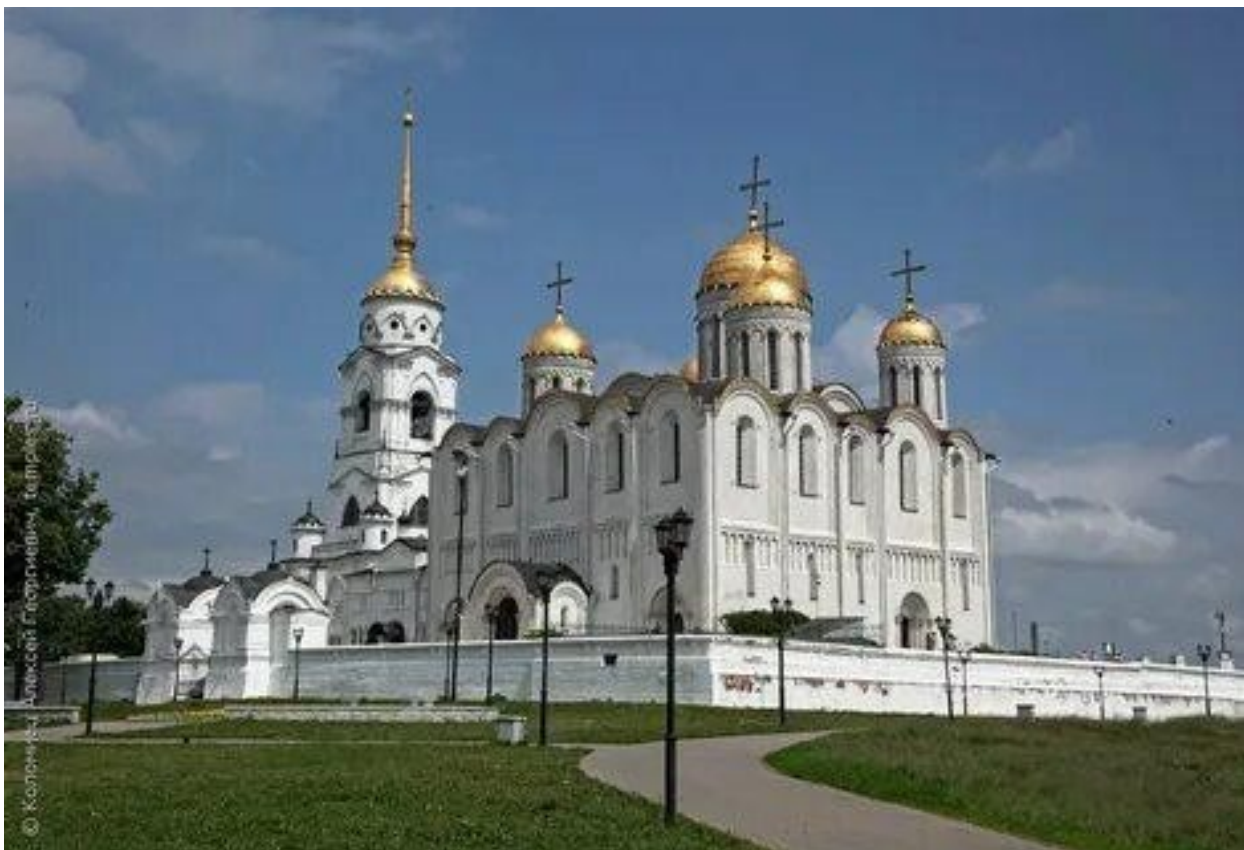
Ярославский дворец молодёжи
there is a building with a sign on the top of it

Для декодирования изображений, закодированных в формате base64, можно использовать следующий код:

```
import base64
import requests
import matplotlib.pyplot as plt
from PIL import Image
import PIL
from io import BytesIO

def get_image(data):
    image = Image.open(BytesIO(base64.b64decode(data))).convert('RGB')
    return image

get_image(data.sample(1)['image'].tolist()[0])
```



В данных есть выбросы — записи, не относящихся к туристическим или историческим объектам. Для очистки данных можно, например, использовать алгоритм TF-IDF: создаем корпус текстов из описаний, применяем TF-IDF, извлекаем наиболее редко или часто встречающиеся слова. Они могут указывать на мусорные данные, такие как мемы, баннеры или личные фото людей. После выделения подозрительных паттернов, можно провести ручную проверку и фильтрацию данных.

Для дальнейшего тестирования и получения чистых сэмплов, рекомендуется взять не менее 100 чистых образцов со стратификацией по городу и названию места. Это можно сделать, выбрав случайным образом несколько образцов из каждой категории (город + название места), чтобы обеспечить равномерное представительство и разнообразие данных.

!! Важно: Помните, что чистые сэмплы одного и того же места выдадут примерно одинаковое описание. Есть несколько стратегий, как повысить разнообразие ответов:

- выбор одного самого лучшего:
 - самое длинное описание
 - самое подходящие по мнению LLM
- конкатенация описаний (но важно не сконкатенировать слишком много)

Изображения можно отдать в мета-данные.

Проверим датасет на предмет пропусков

Здесь предлагаем вам построить RAG с использованием моделей, которые не были представлены на семинаре (включая другие векторные базы данных). Можно воспользоваться материалами по этой ссылке: [data_connection/vectorstores](https://data-connection.com/vectorstores).

!Важно, не забудьте отранжировать чанки текстов с помощью `ragatouille`. На практике порядок текстов в промпте влияет на точность ответа.

Вместо RAGatouille из `ragatouille` буду использовать альтернативу (`FlagReranker`)

```
df = data.drop('en_txt', axis = 1)
df['desc_len'] = df['description'].str.len()
df = df.sort_values('desc_len', ascending=False)
df_grouped = df.groupby('WikiData').agg({
    'Name': lambda x: x.value_counts().idxmax(),
    'City': lambda x: x.value_counts().idxmax(),
    'Lon': 'mean',
    'Lat': 'mean',
    'description': 'first',
    'image': lambda x: list(x),
}).reset_index()

data = df_grouped
data.head()
```

	WikiData	Name	City
\			
0	Q100270892	Нижегородский Кремль	Нижний Новгород
1	Q100701435	Гарнизонная казарма	Нижний Новгород
2	Q100882660	Губернаторский сад	Нижний Новгород
3	Q104554178	Церковь Николы в Меленках	Ярославль
4	Q104776528	Церковь в честь Преображения Господня	Нижний Новгород

	Lon	Lat	
description \			
0	44.005554	56.327255	историческое здание в Нижнем Новгороде, Россия
1	44.004299	56.327614	здание в Нижегородском кремле, Россия
2	44.005272	56.329803	парк в Нижегородском кремле, Россия
3	39.834522	57.609947	Православный храм в Ярославле
4	44.072117	56.318909	Спасо-Преображенская церковь (Старые

Печёры)

```
image
0  [/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...
1  [/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...
2  [/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...
3  [/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...
4  [/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...

# TF-IDF для выявления и удаления выбросов

tfidf = TfidfVectorizer(max_features=2000, ngram_range=(1, 2))

tfidf_matrix = tfidf.fit_transform(data['description'].fillna(''))

avg_tfidf = np.array(tfidf_matrix.mean(axis=1)).flatten()
data['tfidf_score'] = avg_tfidf

# Отсечение (100 - percentile)% текстов с наибольшим TF-IDF скором
percentile = 95
threshold = np.percentile(avg_tfidf, percentile)
data['is_outlier'] = data['tfidf_score'] > threshold

clean_data = data[~data['is_outlier']].copy()

# Стратификация (выбор образцов: Город-Достопримечательность)
clean_data['group_key'] = clean_data['City'].astype(str) + '||' +
clean_data['Name'].astype(str)

stratified_data = clean_data.groupby('group_key',
group_keys=False).apply(
    lambda x: x.sample(min(len(x), 1), random_state=42)
).reset_index(drop=True)

if 'group_key' in stratified_data.columns:
    stratified_data = stratified_data.drop(columns=['group_key'])

print(f'Чистых записей: {len(clean_data)}')
print(f'Удалено выбросов: {data['is_outlier'].sum()}')
print(f'Итоговый размер выборки: {len(stratified_data)}')

Чистых записей: 281
Удалено выбросов: 15
Итоговый размер выборки: 281

stratified_data.sample(5)
```

	WikiData	Name	City
Lon \			
117	Q4165268	Дом Болконского	Ярославль
39.900871			

213	Q4525185	Школа № 33 имени Карла Маркса	Ярославль
39.881363			
66	Q2403914	Спасо-преображенский собор	Нижний Новгород
43.872032			
42	Q18331765	Церковь Никиты Великомученика	Владимир
40.397797			
6	Q107506739	Серго Орджоникидзе	Екатеринбург
60.592449			

	Lat		description \
117	57.625092		Дом Болконского
213	57.628075	Школа № 33 имени Карла Маркса	
66	56.349922	православный храм в Нижнем Новгороде	
42	56.130745	Никитская церковь	
6	56.885479	Памятник в Екатеринбурге	

	image	tfidf_score \
117	[/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...	0.001642
213	[/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...	0.002923
66	[/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...	0.002526
42	[/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...	0.001648
6	[/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUEBAUEAwUFB...	0.001648

	is_outlier
117	False
213	False
66	False
42	False
6	False

```
documents = []
for _, row in stratified_data.iterrows():
    # Объединение текста для получения большей информации
    full_text = f"Название: {row['Name']}, Город: {row['City']},
    Описание: {row['description']}"

    # Создаем документ для RAG
    doc = LangchainDocument(
        page_content=full_text,
        metadata={
            'image': row['image'],
            'description': row['description'],
            'name': row['Name'],
            'city': row['City'],
            'lon': row['Lon'],
            'lat': row['Lat'],
            'wikidata': row['WikiData']
        }
    )
```

```
documents.append(doc)
print(f'Всего записей {len(documents)}')
```

Всего записей 281

Настройка RAG

Создание эмбедингов

```
model_name = 'sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2'
```

```
embeddings = HuggingFaceEmbeddings(
    model_name=model_name,
    model_kwargs={'device': 'cpu'},
    encode_kwargs={'normalize_embeddings': True}
)
```

Создание чанков

```
text_splitter = SpacyTextSplitter(
    pipeline='ru_core_news_sm',
    chunk_size=500,
    chunk_overlap=50
)
```

```
chunks = text_splitter.split_documents(documents)
```

```
print(f'Создано {len(chunks)} чанков из {len(documents)} документов')
print(f'Пример чанка: {chunks[16].page_content}...')
```

Создано 281 чанков из 281 документов

Пример чанка: Название: Мемориальный дом-музей М. Богдановича, Город: Ярославль, Описание: Мемориальный дом-музей М. Богдановича...

Векторное хранилище

```
vectorstore = FAISS.from_documents(
    documents=chunks,
    embedding=embeddings,
    distance_strategy=DistanceStrategy.COSINE)
```

Загрузка модели переранжировщика

```
os.environ['HF_HUB_DISABLE_PROGRESS_BARS'] = '1'
reranker_model = FlagReranker('BAAI/bge-reranker-large',
    use_fp16=True)
```

```
def retrieve_and_rerank(question, k=5, rerank_k=10):
```

```
    """
```

Поиск с переранжированием


```

"""
# Начальный поиск
initial_results = vectorstore.similarity_search(question,
k=rerank_k)

if not initial_results:
    return []

# Переранжирование
pairs = [[question, doc.page_content] for doc in initial_results]
scores = reranker_model.compute_score(pairs)

for doc, score in zip(initial_results, scores):
    doc.metadata['relevance_score'] = float(score)

# Сортировка
sorted_results = sorted(initial_results, key=lambda x:
x.metadata['relevance_score'], reverse=True)

# Формирование результатов
final_results = []
for res in sorted_results[:k]:
    final_results.append({
        'content': res.page_content,
        'detailed': res.metadata.get('detailed_text',
res.page_content),
        'metadata': res.metadata,
        'score': res.metadata['relevance_score']
    })

return final_results

# Проверка

question = 'Какие достопримечательности есть в Нижнем Новгороде?'
answer = retrieve_and_rerank(question)
for i in range(len(answer)):
    print(answer[i]['content'])

```

You're using a XLRobertaTokenizerFast tokenizer. Please note that with a fast tokenizer, using the `__call__` method is faster than using a method to encode the text followed by a call to the `pad` method to get a padded encoding.

Название: Нижегородский Кремль, Город: Нижний Новгород, Описание: историческое здание в Нижнем Новгороде, Россия

Название: Усадьба Вяхиревых, Город: Нижний Новгород, Описание: Усадьба Вяхиревых

Название: Чкаловская лестница, Город: Нижний Новгород, Описание:

лестница на Волжском откосе в Нижнем Новгороде
Название: Усадьба Киришбаумов, Город: Нижний Новгород, Описание:
Усадьба Киришбаумов
Название: Дом А. К. Фомина, Город: Нижний Новгород, Описание: памятник

Выполните визуализацию эмбедингов при помощи метода главных компонент (PCA) и Uniform Manifold Approximation and Projection (UMAP). Пожалуйста, выполните PCA независимо от расстар.

```
# Список текстов и список эмбедингов
texts = [chunk.page_content for chunk in chunks]
embeddings_list = embeddings.embed_documents(texts)

# Извлекаем города из метаданных
cities = [chunk.metadata.get('city', 'Unknown') for chunk in chunks]

# Палитра цветов для городов
unique_cities = list(set(cities))
color_palette = ['red', 'blue', 'green', 'orange', 'purple']
city_to_color = {city: color_palette[i % len(color_palette)] for i,
city in enumerate(unique_cities)}

# Масштабирование эмбедингов для PCA

scaler = StandardScaler()
sample_embeddings_scaled = scaler.fit_transform(embeddings_list)

# Настройка PCA
pca = PCA(n_components=2)
pca_result = pca.fit_transform(sample_embeddings_scaled)
explained_variance = pca.explained_variance_ratio_

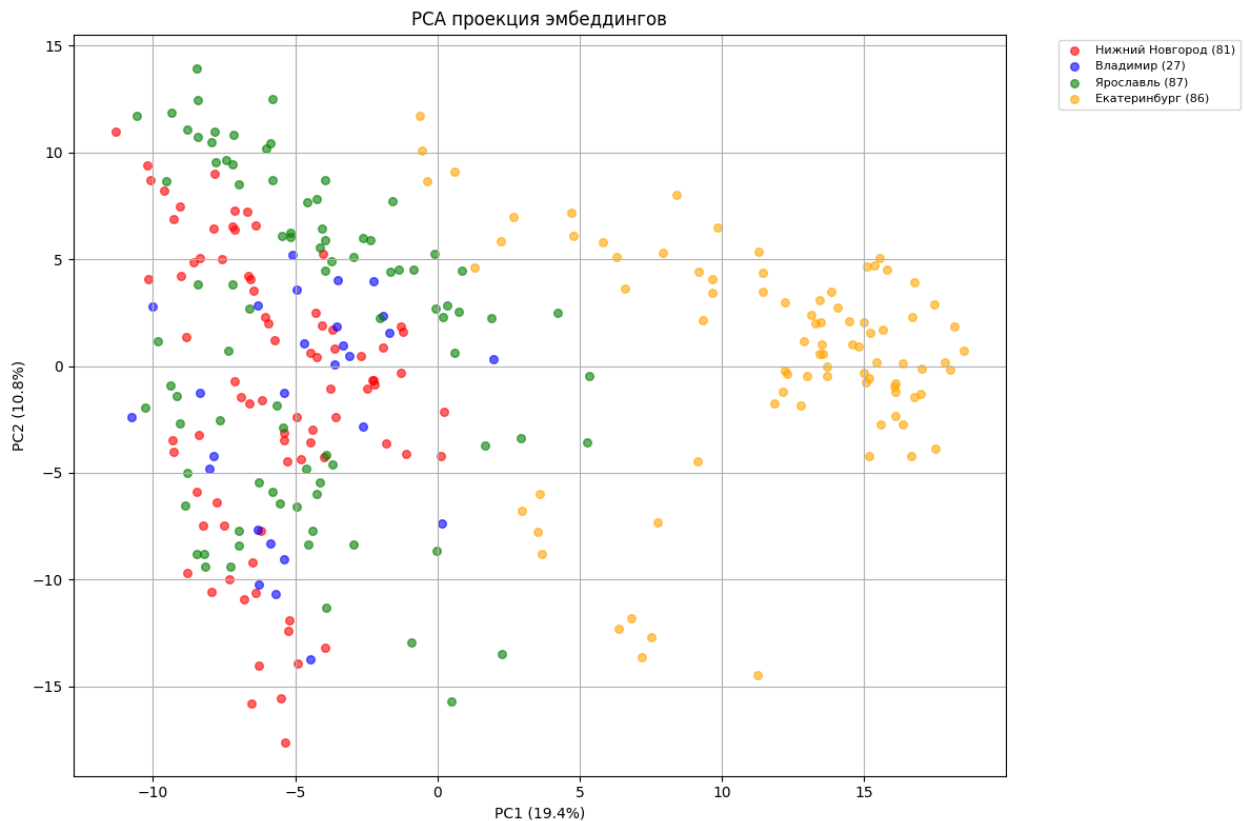
# Визуализация PCA
plt.figure(figsize=(12, 8))

for city in unique_cities:
    # Маска для текущего города во всем наборе данных
    mask = [c == city for c in cities]
    if sum(mask) == 0: # пропуск, если город не найден
        continue

    plt.scatter(
        pca_result[mask, 0],
        pca_result[mask, 1],
        label=f'{city} ({sum(mask)})',
        alpha=0.6,
        s=30,
        color=city_to_color[city]
    )
```

```
plt.title('PCA проекция эмбедингов', fontsize=12)
plt.xlabel(f'PC1 ({explained_variance[0]:.1%})', fontsize=10)
plt.ylabel(f'PC2 ({explained_variance[1]:.1%})', fontsize=10)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', fontsize=8)
plt.grid()
plt.tight_layout()

plt.show()
```



Масштабирование эмбедингов для UMAP

```
scaler_mm = MinMaxScaler()
sample_embeddings_mm = scaler_mm.fit_transform(embeddings_list)
```

Настройка UMAP

```
umap_reducer = umap.UMAP(
    n_components=2,
    n_neighbors=15,
    min_dist=0.1,
    metric='euclidean',
    random_state=42
)
umap_result = umap_reducer.fit_transform(sample_embeddings_mm)
```

```
# Визуализация UMAP
plt.figure(figsize=(12, 8))

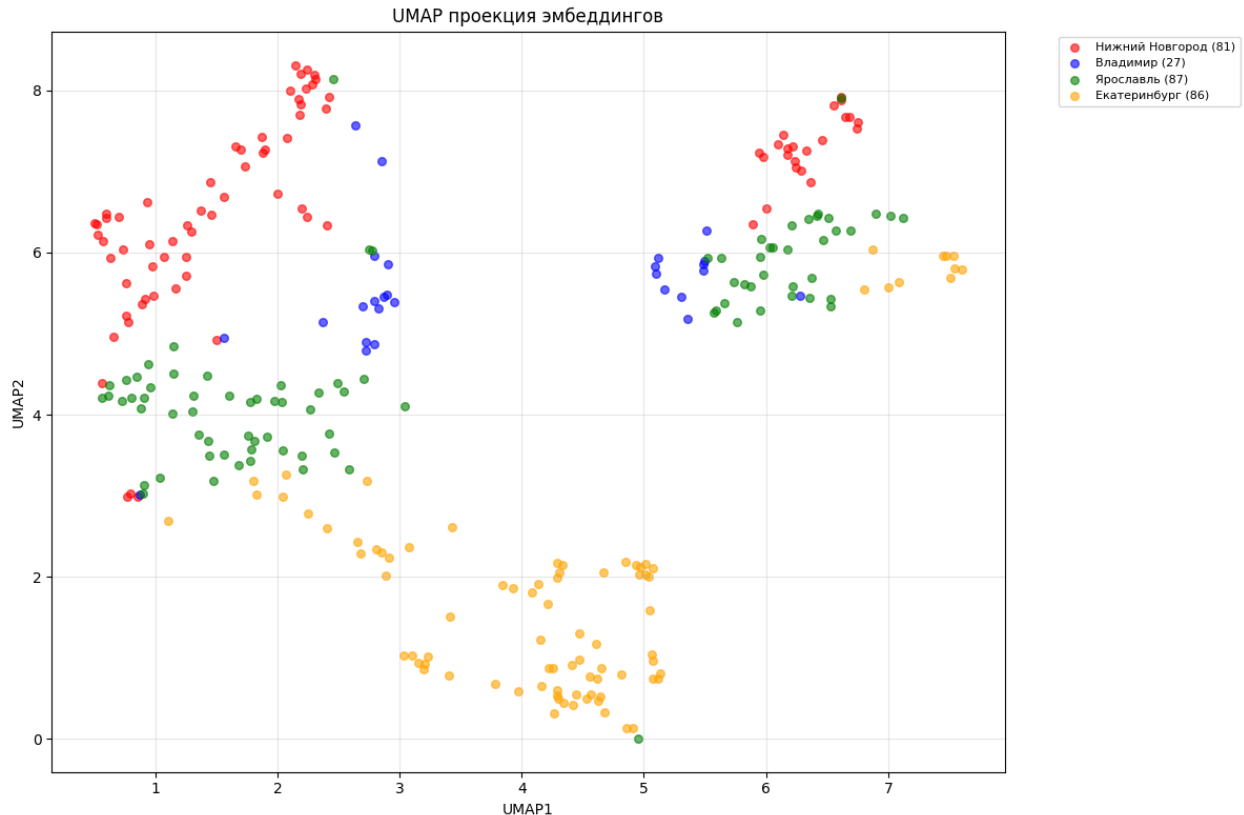
for city in unique_cities:
    # Маска для текущего города во всем наборе данных
    mask = [c == city for c in cities]
    if sum(mask) == 0: # пропуск, если город не найден
        continue

    plt.scatter(
        umap_result[mask, 0],
        umap_result[mask, 1],
        label=f'{city} ({sum(mask)}',
        alpha=0.6,
        s=30,
        color=city_to_color[city]
    )

plt.title('UMAP проекция эмбедингов', fontsize=12)
plt.xlabel('UMAP1', fontsize=10)
plt.ylabel('UMAP2', fontsize=10)
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left', fontsize=8)
plt.grid(True, alpha=0.3)

plt.tight_layout()

plt.show()
```



RAGAS (3 балла)

RAGAS - это система, созданная для автоматической оценки эффективности модели RAG. Она представляет собой комплексный инструмент, способный оценить как качество извлечения информации, так и качество созданного текста. Это достигается благодаря использованию различных метрик, которые анализируют разные аспекты ответов, сгенерированных языковой моделью.

Метрики оценки в RAGAS

Faithfulness

Оценивает, насколько ответы языковой модели верно передают информацию, содержащуюся в источниках данных. Это критически важно, чтобы гарантировать, что модель не искажает или не меняет информацию при ее интерпретации.

$$\text{Faithfulness} = \frac{\text{Number of claims that can be inferred from given context}}{\text{Total number of claims in the generated answer}}$$

Answer Relevancy

Демонстрирует, насколько ответы соответствуют заданным вопросам. Это важный аспект, так как даже правильный ответ с технической точки зрения может быть ненужным, если он не отвечает на конкретный вопрос пользователя.

$$\text{answer relevancy} = \frac{1}{N} \sum_{i=1}^N \cos(E_{g_i}, E_o)$$

$$\text{answer relevancy} = \frac{1}{N} \sum_{i=1}^N \frac{E_{g_i} \cdot E_o}{\|E_{g_i}\| \|E_o\|}$$

Где:

- E_{g_i} - это эмбединг сгенерированного ответа i
- E_o - это эмбединг исходного вопроса.
- N - это количество сгенерированных вопросов, которое по умолчанию равно 3.

Context Recall и Context Precision

Эти метрики измеряют эффективность модели в учете контекста вопроса для формирования ответа. "Context Recall" оценивает количество использованной информации из контекста в ответе, в то время как "Context Precision" оценивает точность и релевантность использования контекстной информации.

- $\text{context recall} = \frac{\text{Number of sentences that can be attributed to context}}{\text{Number of sentences in GT}}$
- $\text{Context Precision}@k = \frac{\sum \text{precision}@k}{\text{total number of relevant items in the top K results}}$
– $\text{Precision}@k = \frac{\text{true positives}@k}{(\text{true positives}@k + \text{false positives}@k)}$

Answer Semantic Similarity

Оценивает степень семантической близости между ответом модели и эталонным ответом. Это позволяет определить, насколько естественно и точно модель способна воссоздавать человеческий язык.

Answer Correctness

Оценка корректности ответа. Это последний шаг в проверке, чтобы убедиться, предоставляет ли модель точную и достоверную информацию в своем ответе.

В этом задании вам нужно реализовать метрику `answer_relevancy` в коде. Опционально за допбаллы можно реализовать другие метрики.

Для этого датасет нужно привести к виду, в котором будут следующие поля:

1. question

- Смысл: Эта колонка содержит вопросы, на которые нужно найти ответы. В контексте задач QA, это основная информация, которую модель должна обработать для генерации ответа.
- Пример: "Каковы преимущества RAG?"

2. ground_truths

- Смысл: В этой колонке находятся правильные ответы на соответствующие вопросы. Это может использоваться для оценки производительности модели, так как позволяет сравнить сгенерированные ответы с эталонными.
- Пример: "RAG улучшает качество генерации, используя внешние источники информации."

3. answer

- Смысл: Эта колонка содержит ответы, которые генерирует модель на основе заданных вопросов. Это может быть результат работы модели, которую вы тестируете или обучаете.
- Пример: "RAG позволяет моделям извлекать информацию из внешних источников, что улучшает их способность отвечать на сложные вопросы."

4. contexts

- Смысл: Эта колонка предоставляет контекст или дополнительную информацию, которая может быть полезна для понимания вопроса или для формирования ответа. Это может быть текст, из которого был извлечен ответ, или другие релевантные документы.
- Пример: "RAG (Retrieval-Augmented Generation) — это метод, который использует внешние базы данных для получения информации при генерации текста."

HINT: Поле `question` можно тоже сгенерировать. Например, можно попросить модель задать вопрос по описанию изображения.

HINT: Для `answer_relevancy` нужны не все поля

HINT: Инференс такого эксперимента может быть долгим. План минимум: возьмите 100 случайных сэмплов.

```
def make_prompt(results, question):  
    """  
    Создание промпта для модели  
    """  
    main_res = results[0]  
    other_res = results[1:]
```

```

main_metadata = main_res['metadata']
main_context = f"""
{main_metadata.get('name', '')} находится в городе
{main_metadata.get('city', '')}.
Это {main_metadata.get('description', '')}.
Он расположен на широте {main_metadata.get('lat', '')} и долготе
{main_metadata.get('lon', '')}.
Более подробную информацию можно узнать на викидате, ID -
{main_metadata.get('wikidata', '')}.
"""

other_data = [f"{x['metadata'].get('name', '')} в городе
{x['metadata'].get('city', '')}"
              for x in other_res]
other_context = ' '.join(other_data)

result_prompt = f"""
<|im_start|>system
Ты помощник туриста. Отвечай естественно, как человек, на русском
языке.
<|im_end|>
<|im_start|>user
Контекст: {main_context}
Дополнительные данные: {other_context}
Вопрос: {question}

Используй только информацию из контекста. Если не знаешь - честно
признайся, но предложи помощь в другом вопросе.
<|im_end|>
<|im_start|>assistant
"""

return result_prompt, main_context, other_context

quant_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_compute_dtype=torch.float16
)

model_answer_name = "Qwen/Qwen2.5-3B-Instruct"

tokenizer = AutoTokenizer.from_pretrained(model_answer_name)
tokenizer.pad_token = tokenizer.eos_token

model_answer = AutoModelForCausalLM.from_pretrained(
    model_answer_name,
    torch_dtype=torch.float16,
    device_map="auto"
)

```


Sliding Window Attention is enabled but not implemented for `sdpa`; unexpected results may be encountered.

```
{"model_id": "4098e3a95e094a1b984db52a9c589d36", "version_major": 2, "version_minor": 0}
```

```
def get_answer(question, k=3):
    results = retrieve_and_rerank(question, k=k)
    prompt, main_context, other_context = make_prompt(results,
question)

    inputs = tokenizer(prompt, return_tensors="pt", truncation=True,
max_length=1500).to(model_answer.device)

    outputs = model_answer.generate(
        **inputs,
        max_new_tokens=250,
        do_sample=True,
        temperature=0.3,
        top_p=0.9,
        repetition_penalty=1.1,
        pad_token_id=tokenizer.eos_token_id
    )

    full_text = tokenizer.decode(outputs[0], skip_special_tokens=True)

    assistant_marker = "<|im_start|>assistant"
    if assistant_marker in full_text:
        answer = full_text.split(assistant_marker)[-1].strip()
    else:

        parts = full_text.split("assistant")
        answer = parts[-1].strip() if len(parts) > 1 else full_text

    context = f'''
Информация о самом релевантном объекте: {main_context}
Дополнительно: {other_context}
'''

    images = results[0]['metadata']['image'] if results else []

    del inputs, outputs
    torch.cuda.empty_cache()

    return answer, context, images

question = 'Расскажи про Нижегородский Кремль'

answer, context, images = get_answer(question)
print(answer)
```

```
for img in images[:5]:  
    display(get_image(img))
```

Нижегородский Кремль – это историческое здание, расположенное в Нижнем Новгороде, России. Его географические координаты – широта 56.327255 и долгота 44.005554.

Этот кремль является важным архитектурным памятником и представляет собой часть города с историческим значением. Однако у меня нет дополнительной информации о конкретных достопримечательностях внутри кремля, таких как Георгиевская башня или Гарнизонная казарма. Для получения более детальной информации рекомендую обратиться к официальным источникам или посетить сайт википедии с указанным ID Q100270892 для получения полного списка доступных данных.

Если вам нужна информация о других достопримечательностях Нижнего Новгорода, я буду рад помочь!







```
def answer_relevancy(question=None, answer=None,
embedding_model=embeddings):

    # Эмбединги
    question_emb = embedding_model.embed_query(question)
    answer_emb = embedding_model.embed_query(answer)

    # Подготовка для анализа через cosine_similarity
    question_emb_array = np.array(question_emb).reshape(1, -1)
    answer_emb_array = np.array(answer_emb).reshape(1, -1)

    # Косинусное сходство
    similarity = cosine_similarity(question_emb_array,
answer_emb_array)[0][0]

    # Штраф за слишком короткие ответы
    length_penalty = min(1.0, len(answer.split()) / 30)

    return similarity * length_penalty
```

Протестируйте ваш RAG (3 балла)

Создадим по 4-6 вопросов разной категории. Сохраним категорию в метаданные для дальнейшего анализа.

```
test_questions = []
test_metadata = []

# Создание списков с уникальными городами и достопримечательностями
unique_cities = stratified_data['City'].unique()
unique_places = stratified_data[['Name', 'City']].drop_duplicates()

# Вопросы о том, какие достопримечательности есть в городе
for city in unique_cities:
    test_questions.append(f'Какие достопримечательности или интересные
места есть в городе {city}?')
    test_metadata.append({'type': 'Достопримечательности в городе',
'city': city, 'name': None})

# Вопросы о конкретной достопримечательности
sample_places = unique_places.sample(n=6, random_state=42)
for _, row in sample_places.iterrows():
    test_questions.append(f'Расскажите о {row["Name"]} в городе
{row["City"]}')
    test_metadata.append({'type': 'Описание', 'city': row['City'],
'name': row['Name']})

# Вопросы о том, какие музеи есть в городе
```

```

for city in unique_cities:
    test_questions.append(f'Какие музеи есть в городе {city}?')
    test_metadata.append({'type': 'Музеи в городе', 'city': city,
'name': None})

# Вопросы о местоположении достопримечательности
sample_places = unique_places.sample(n=6, random_state=4721)
for _, row in sample_places.iterrows():
    test_questions.append(f'В каком городе находится {row["Name"]}?')
    test_metadata.append({'type': 'Город по достопримечательности',
'city': row['City'], 'name': row['Name']})

# Вопросы о ближайших достопримечательностях
sample_places_nearby = unique_places.sample(n=5, random_state=7956)
for _, row in sample_places_nearby.iterrows():
    test_questions.append(f'Что находится рядом с {row["Name"]} в
городе {row["City"]}?')
    test_metadata.append({'type': 'Ближайшие достопримечательности или
интересные места', 'city': row['City'], 'name': row['Name']})

print(f'Создано тестовых вопросов: {len(test_questions)}')

```

Создано тестовых вопросов: 25

Результаты тестирования на 10 случайных вопросах

```

sample_questions = random.sample(test_questions, 10)
for question in sample_questions:
    print(f'\nВопрос: {question}')
    answer, context, images = get_answer(question)
    print(f'Ответ: {answer}')
    relevancy = answer_relevancy(question, answer)
    print(f'Answer Relevancy: {relevancy:.4f}')

```

Вопрос: В каком городе находится Северная башня?

Ответ: Северная башня находится в городе Нижний Новгород.

Answer Relevancy: 0.1420

Вопрос: Какие достопримечательности или интересные места есть в городе Владимир?

Ответ: В городе Владимир находятся несколько интересных достопримечательностей:

1. ****Бывшая водонапорная башня**** – это исторический объект, который стоит посетить для того, чтобы увидеть его строение и узнать больше о истории города.
2. ****Владимирский центральный**** – возможно, это вокзал или станция железнодорожного сообщения, так как часто называют городские вокзалы "центрами".

3. ****Золотые ворота**** – это архитектурный памятник, который также стоит посмотреть.

Если вам нужна более детальная информация о каждом из этих мест, вы можете найти дополнительные сведения на википедии по соответствующим ID-номерам (Q4114100 для башни).

Если у вас есть еще вопросы или нужно что-то конкретное узнать, дайте знать!

Answer Relevancy: 0.8118

Вопрос: Какие музеи есть в городе Екатеринбург?

Ответ: В городе Екатеринбург есть один музей, который я могу назвать по информации, которую мне предоставили:

1. Музей истории Екатеринбурга

Если вы хотите узнать больше о нем или других музеях города, рекомендую обратиться к официальным сайтам музеев или использовать викидату для получения дополнительной информации.

К сожалению, из предоставленной информации я не могу указать другие музеи в Екатеринбурге. Возможно, вам стоит проверить еще раз список музеев, которые вы упомянули (Художественный Музей Эрнста Неизвестного и Екатеринбургский музей изобразительных искусств), чтобы убедиться, что они действительно находятся в Екатеринбурге.

Если у вас есть конкретный интересующий вас музей, например, Художественный Музей Эрнста Неизвестного, я бы мог попробовать найти более детальную информацию об этом месте. Но основная информация о других музеях в

Answer Relevancy: 0.9001

Вопрос: Какие достопримечательности или интересные места есть в городе Екатеринбург?

Ответ: В Екатеринбурге есть несколько интересных мест:

1. Музей истории Екатеринбурга – это уникальное место для знакомства с историей города.

2. Здание городской электростанции "Луч" – это красивое здание, которое можно посетить и сфотографироваться у его стен.

Если вам нужна более детальная информация о этих объектах или другие достопримечательности, дайте знать! Я помогу найти дополнительные факты.

Но помните, что викидата (Q4306294) предоставляет больше информации об этом музее, если вы захотите узнать все детали.

Answer Relevancy: 0.8483

Вопрос: Расскажите о Владимирская духовная семинария в городе Владимир

Ответ: Владимирская духовная семинария – это учебное заведение для подготовки будущих священников, расположенная в городе Владимир. Она находится на координатах широты 56.131168 и долготы 40.413074.

Для получения более детальной информации вы можете посетить викидату с указанным ID Q4113251.

Также в городе Владимир есть другие интересные места:

- Владимирское духовное училище,
- Костел Святого Розария Пресвятой Девы Марии Римско-Католической Церкви.

Если вам нужно больше информации об этих объектах или других достопримечательностях города Владимир, я буду рад помочь!

Answer Relevancy: 0.8805

Вопрос: В каком городе находится Беседка на Волжской набережной?

Ответ: Беседка на Волжской набережной находится в городе Ярославль.

Answer Relevancy: 0.2047

Вопрос: Какие музеи есть в городе Нижний Новгород?

Ответ: К сожалению, в предоставленном контексте нет информации о музеях в городе Нижний Новгород. Информация о музеях в Нижнем Новгороде отсутствует. Возможно, вы могли бы уточнить запрос или предоставить дополнительный контекст?

Если вам нужно найти информацию о музеях в других городах, я буду рад помочь! Например, если вы хотите узнать об истории музея древнерусского искусства в Ярославле, я могу это сделать.

Answer Relevancy: 0.8531

Вопрос: Какие музеи есть в городе Владимир?

Ответ: В городе Владимир есть исторический музей. Также рядом с городом находится Владимиро-Суздальский музей-заповедник, который является центральным офисом этого музея. Кроме того, в Ярославле находится музей древнерусского искусства.

Если вам нужна более детальная информация или адреса этих музеев, я могу попробовать найти дополнительные данные для вас. Но основная информация, которую вы запросили, уже предоставлена в вашем исходном тексте.

Answer Relevancy: 0.8973

Вопрос: Расскажите о Дом Е. Трушенинниковой в городе Нижний Новгород

Ответ: Дом Евгении Трушенинниковой в Нижнем Новгороде – это особняк, созданный известным архитектором. Он расположен по адресу с указанными координатами (широта 56.329013999999994, долгота 44.010155) в

историческом центре города.

Хотя конкретные детали об этом здании не приведены в предоставленной информации, дом обычно считается значимым объектом культурного наследия города. Архитектура такого дома может быть интересна для туристов благодаря своему стилю и эпохе его строительства.

Если вам нужна более точная информация или вы хотите узнать больше о доме, рекомендую обратиться к специализированным источникам, возможно, на викидайте с указанием ID Q84172585. Также можно попробовать найти дополнительную информацию на сайте музеев или архитектурных фондов

Answer Relevancy: 0.7871

Вопрос: Какие достопримечательности или интересные места есть в городе Ярославль?

Ответ: К сожалению, в предоставленной информации нет никакой информации о достопримечательностях города Ярославль. Все упомянутые объекты находятся в других городах: усадьба Расторгуевых-Харитоновых в Екатеринбурге, а также усадьба Вяхиревых в Нижнем Новгороде и Волжская башня в Ярославле.

Если вам нужна дополнительная информация о достопримечательностях Ярославля, я могу попробовать найти это самостоятельно или посоветовать другие источники для получения такой информации.

Answer Relevancy: 0.8270

Дополнительная часть (+3 балла)

Реализуйте одну из метрик retrieval-части на выбор.

Для того, чтобы понять model as a judge подход из RAGAS, советуем посетить [этот кукбук](#)

Посмотрим на вопросы тестового списка и посчитаем общие метрики и релевантность по категориям.

Оценка Answer Relevancy

Посмотрим на тестовые вопросы, сгенерированные ранее

```
for i, question in enumerate(test_questions, start=1):  
    print(f'{i}. {question}')
```

1. Какие достопримечательности или интересные места есть в городе Нижний Новгород?
2. Какие достопримечательности или интересные места есть в городе Ярославль?
3. Какие достопримечательности или интересные места есть в городе Екатеринбург?
4. Какие достопримечательности или интересные места есть в городе

Владимир?

5. Расскажите о Церковь Александра Невского в Сормово в городе Нижний Новгород
6. Расскажите о Колокольня Никитского прихода в городе Ярославль
7. Расскажите о Храм на Крови в городе Екатеринбург
8. Расскажите о Дом Е. Трушенинниковой в городе Нижний Новгород
9. Расскажите о Владимирская духовная семинария в городе Владимир
10. Расскажите о Комсомолу Урала в городе Екатеринбург
11. Какие музеи есть в городе Нижний Новгород?
12. Какие музеи есть в городе Ярославль?
13. Какие музеи есть в городе Екатеринбург?
14. Какие музеи есть в городе Владимир?
15. В каком городе находится Музей истории, науки и техники Свердловской железной дороги?
16. В каком городе находится Церковь Александра Невского в Сормово?
17. В каком городе находится Беседка на Волжской набережной?
18. В каком городе находится Дом А. И. Фролова?
19. В каком городе находится Суворовское училище?
20. В каком городе находится Северная башня?
21. Что находится рядом с улица Чернышевского в городе Екатеринбург?
22. Что находится рядом с Дом контор в городе Екатеринбург?
23. Что находится рядом с Парк имени Кулибина в городе Нижний Новгород?
24. Что находится рядом с Театр кукол в городе Екатеринбург?
25. Что находится рядом с Свердловский областной краеведческий музей в городе Екатеринбург?

Будем оценивать общую релевантность и релевантность по типу вопроса

```
# Словарь для хранения результатов по типам вопросов
results_by_type = {}

# Оценка для каждого вопроса с сохранением типа вопроса
for i, (question, metadata) in enumerate(tqdm(zip(test_questions,
test_metadata),
desc='Оценка релевантности',
total=len(test_questions))):

    answer, context, images = get_answer(question)
    relevancy = answer_relevancy(question, answer)

    question_type = metadata['type']

    # Инициализация типа в словаре, если его еще нет
    if question_type not in results_by_type:
        results_by_type[question_type] = {
            'scores': [],
            'questions': [],
            'answers': [],
```

```

        'cities': [],
        'names': []
    }

    # Сохранение результатов
    results_by_type[question_type]['scores'].append(relevancy)
    results_by_type[question_type]['questions'].append(question)
    results_by_type[question_type]['answers'].append(answer)
    results_by_type[question_type]['cities'].append(metadata['city'])
    if metadata['name']:
        results_by_type[question_type]
        ['names'].append(metadata['name'])

    torch.cuda.empty_cache()

{"model_id": "9cceb7e9c5be4ab4ac1b81bd6ca53e41", "version_major": 2, "version_minor": 0}

# Общая статистика

all_scores = []
for q_type, data in results_by_type.items():
    all_scores.extend(data['scores'])

avg_all_scores = np.mean(all_scores)
min_all_scores = np.min(all_scores)
max_all_scores = np.max(all_scores)

print(f'Всего вопросов: {len(all_scores)}')
print(f'Средняя Answer Relevancy: {avg_all_scores:.4f}')
print(f'Минимум Answer Relevancy: {min_all_scores:.4f}')
print(f'Максимум Answer Relevancy: {max_all_scores:.4f}')

Всего вопросов: 25
Средняя Answer Relevancy: 0.7007
Минимум Answer Relevancy: 0.1255
Максимум Answer Relevancy: 0.9109

# Статистика по каждому типу

for q_type, data in results_by_type.items():
    scores = data['scores']
    avg_score = np.mean(scores)
    min_score = np.min(scores)
    max_score = np.max(scores)

    print(f'\n{q_type.upper()}:')
    print(f'Всего вопросов: {len(scores)}')
    print(f'Средняя Answer Relevancy: {avg_score:.4f}')

```

```
print(f'Минимум Answer Relevancy: {min_score:.4f}')
```

```
print(f'Максимум Answer Relevancy: {max_score:.4f}')
```

ДОСТОПРИМЕЧАТЕЛЬНОСТИ В ГОРОДЕ:

Всего вопросов: 4
Средняя Answer Relevancy: 0.8553
Минимум Answer Relevancy: 0.8289
Максимум Answer Relevancy: 0.8978

ОПИСАНИЕ:

Всего вопросов: 6
Средняя Answer Relevancy: 0.7981
Минимум Answer Relevancy: 0.6990
Максимум Answer Relevancy: 0.8822

МУЗЕИ В ГОРОДЕ:

Всего вопросов: 4
Средняя Answer Relevancy: 0.8652
Минимум Answer Relevancy: 0.8348
Максимум Answer Relevancy: 0.9109

ГОРОД ПО ДОСТОПРИМЕЧАТЕЛЬНОСТИ:

Всего вопросов: 6
Средняя Answer Relevancy: 0.2777
Минимум Answer Relevancy: 0.1255
Максимум Answer Relevancy: 0.7545

БЛИЖАЙШИЕ ДОСТОПРИМЕЧАТЕЛЬНОСТИ ИЛИ ИНТЕРЕСНЫЕ МЕСТА:

Всего вопросов: 5
Средняя Answer Relevancy: 0.8362
Минимум Answer Relevancy: 0.7709
Максимум Answer Relevancy: 0.8828

Лучший и худший ответ на вопрос среди всех ответов

```
all_scores = []  
all_questions = []  
all_answers = []  
  
for q_type, data in results_by_type.items():  
    all_scores.extend(data['scores'])  
    all_questions.extend(data['questions'])  
    all_answers.extend(data['answers'])
```

```
all_scores = np.array(all_scores)  
all_questions = np.array(all_questions)  
all_answers = np.array(all_answers)
```

Индексы лучшего и худшего ответа

```
best_idx = np.argmax(all_scores)
```

```
worst_idx = np.argmin(all_scores)

print(f'\nЛучший Answer Relevancy: {all_scores[best_idx]:.4f}:')
print(f'Вопрос: {all_questions[best_idx]}')
print(f'Ответ: {all_answers[best_idx]}')

print(f'\nХудший Answer Relevancy: {all_scores[worst_idx]:.4f}:')
print(f'Вопрос: {all_questions[worst_idx]}')
print(f'Ответ: {all_answers[worst_idx]}')
```

Лучший Answer Relevancy: 0.9109:
 Вопрос: Какие музеи есть в городе Екатеринбург?
 Ответ: В городе Екатеринбург есть несколько музеев:

1. Музей истории Екатеринбурга
2. Художественный Музей Эрнста Неизвестного

Если вам нужно больше информации о этих музеях или их расположении, вы можете посмотреть дополнительные данные в викидатах, которые указаны в контексте.

Худший Answer Relevancy: 0.1255):
 Вопрос: В каком городе находится Суворовское училище?
 Ответ: Суворовское училище находится в городе Екатеринбург.

Оценка Context Recall

Промт для оценки полноты контекста

```
def make_prompt_for_recall(model_answer, context):
    return f'''
    <|im_start|>system
    Ты должен оценить, сколько данных из контекста было использовано в
    ответе.
    <|im_end|>
    <|im_start|>user
    Контекст: {context}
    Ответ для оценки: {model_answer}

    Критерии оценки:
    - 1.0: Использована вся информация из контекста (все ключевые
    факты, названия, детали)
    - 0.8: Использована большая часть информации (>70%)
    - 0.6: Использована половина информации (~50%)
    - 0.4: Использована меньшая часть информации (<30%)
    - 0.2: Использовано минимум информации (только отдельные
    фрагменты)
    - 0.0: Информация из контекста не использована или ответ говорит
    "я не знаю"
```

```
Учитывай только информацию из контекста
Ответь только числом от 0.0 до 1.0, без пояснений
<|im_end|>
<|im_start|>assistant
'''
```

Оценка Context Recall

```
def context_recall(context, answer):
    prompt = make_prompt_for_recall(answer, context)
    inputs = tokenizer(prompt,
return_tensors="pt").to(model_answer.device)
    outputs = model_answer.generate(
        **inputs,
        max_new_tokens=10,
        do_sample=True,
        temperature=0.1,
        pad_token_id=tokenizer.eos_token_id
    )
    fullanswer = tokenizer.decode(outputs[0],
skip_special_tokens=True)
    result = fullanswer.split('assistant')[-1].strip()

    score = float(result)
    if 0 <= score <= 1:
        return score
    else:
        return 0.0

recall_scores = []
for idx, question in enumerate(tqdm(test_questions, desc='Оценка
полноты контекста')):
    answer, context, images = get_answer(question)
    score = context_recall(context, answer)
    recall_scores.append(score)
    torch.cuda.empty_cache()

avg_recall_scores = np.mean(recall_scores)
min_recall_scores = np.min(recall_scores)
max_recall_scores = np.max(recall_scores)

print(f'Всего вопросов: {len(recall_scores)}')
print(f'Средняя Context Recall: {avg_recall_scores:.4f}')
print(f'Минимум Context Recall: {min_recall_scores:.4f}')
print(f'Максимум Context Recall: {max_recall_scores:.4f}')

{"model_id": "3f155e70d4574d7f84ac8899068c9c96", "version_major": 2, "vers
ion_minor": 0}
```

Всего вопросов:	25
Средняя Context Recall:	0.6960
Минимум Context Recall:	0.2000
Максимум Context Recall:	0.8000